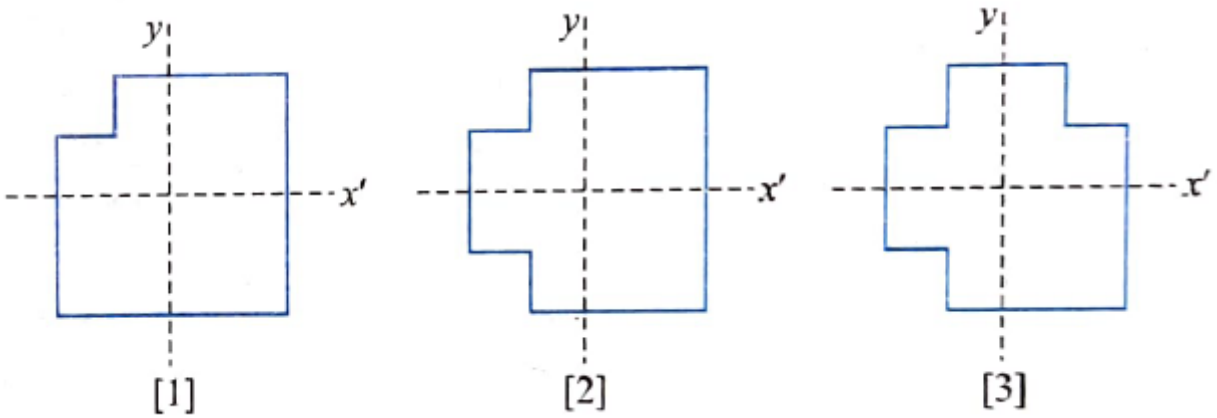
# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

A machinist starts with three identical square plates but cuts one corner from first, two corners from the second and three corners from the third. Rank the three according to the $x$-coordinate of their centre of mass, from smallest to largest.




**(A)** $3, 1, 2$

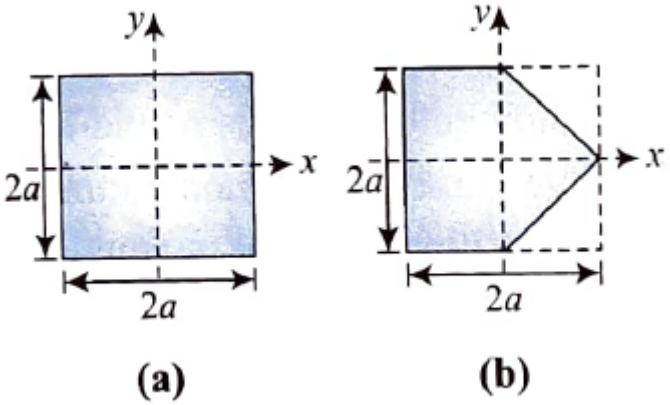
**(B)** $1, 3, 2$

**(C)** $3, 2, 1$

**(D)** 1 and 3 tie, then 2

**Q.2**

Two corners of a square plate as shown in Fig. (a) are folded to make it appear like Fig. (b). Both corners are put together at centre of square ' $O$ '. If $O$ is taken to be origin $(0, 0)$, the centre of mass of Fig. (b) will be at




**(A)** $\left(-\frac{a}{8}, 0\right)$

**(B)** $\left(-\frac{a}{6}, 0\right)$

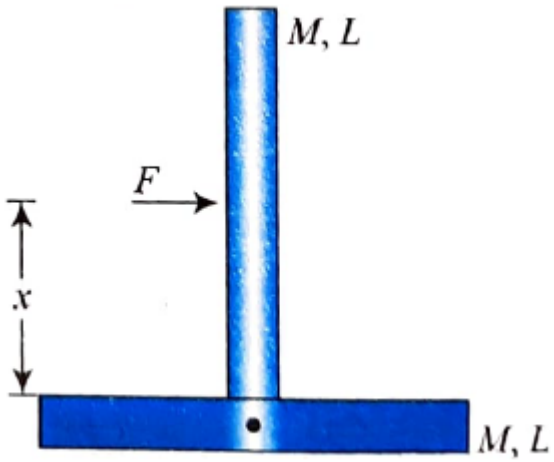
**(C)** $\left(\frac{a}{12}, 0\right)$

**(D)** $\left(-\frac{a}{12}, 0\right)$

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**Q.3**

An inverted $T$-shaped object is placed on a smooth horizontal floor as shown in figure.




A force $F$ is applied on the system as shown in figure. The value of $x$ so that the system performs pure translational motion is

**(A)** $\frac{L}{4}$

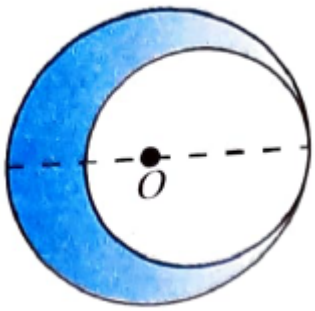
**(B)** $\frac{3L}{4}$

**(C)** $\frac{L}{2}$

**(D)** $\frac{3L}{2}$

**Q.4**

A circular plate of uniform thickness has a diameter of $28\ \text{cm}$. A circular portion of diameter $21\ \text{cm}$ is removed from the plate as shown. $O$ is the centre of mass of complete plate. The position of centre of mass of remaining portion will shift towards left from ' $O$ ' by




**(A)** $5\ \text{cm}$

**(B)** $9\ \text{cm}$

**(C)** $4.5\ \text{cm}$

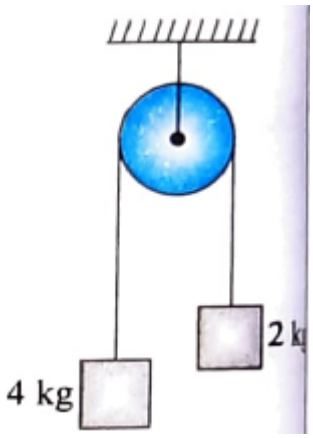
**(D)** $5.5\ \text{cm}$

**Q.5**

For the system shown in figure, the string is light and pulley is frictionless. The $4\ \text{kg}$ block is given an upward velocity of $1\ \text{m/s}$

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The centre of mass of the two blocks will [neglect the impulse duration]




**(A)** accelerate down with $g/3$.

**(B)** initially accelerate downwards with $g$ and then after some time accelerate down with $g/3$.

**(C)** initially accelerate with $\vec{g}$ and then the acceleration is 0

**(D)** initially accelerate with $\vec{g}$ and then with accelerate with $g/9$

**Q.6**

A particle of mass $2\ \mathrm{kg}$ moving with a velocity of $3\ \mathrm{m/s}$ is acted upon by a force which changes its direction of motion by an angle of $90^\circ$ without changing its speed. What is the magnitude of impulse experienced by the particle?

**(A)** $6\ \mathrm{N\ s}$

**(B)** $2\ \mathrm{N\ s}$

**(C)** $3\sqrt{2}\ \mathrm{N\ s}$

**(D)** $6\sqrt{2}\ \mathrm{N\ s}$

**Q.7**

A stationary body explodes into four identical fragment such that three of them fly off mutually perpendicular to each other, each with same $\mathrm{KE}$, $E_0$. The energy of explosion will be

**(A)** $6E_0$

**(B)** $\frac{4E_0}{3}$

**(C)** $4E_0$

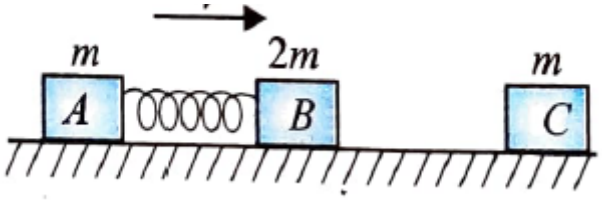
**(D)** $8E_0$

**Q.8**

Two blocks $A$ and $B$ of masses $m$ and $2m$, respectively, are connected with the help of a spring having spring constant, $k$ as shown in figure. Initially, both the blocks are moving with same velocity $v$ on a smooth horizontal plane with the spring in its natural length. During their course of motion, block $B$ makes an inelastic collision with block $C$ of mass $m$ which is initially at rest. The coefficient of restitution for the collision is $1/2$. The maximum compression in the spring is

$v$

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**(A)** $\sqrt{\frac{m}{k}}v$

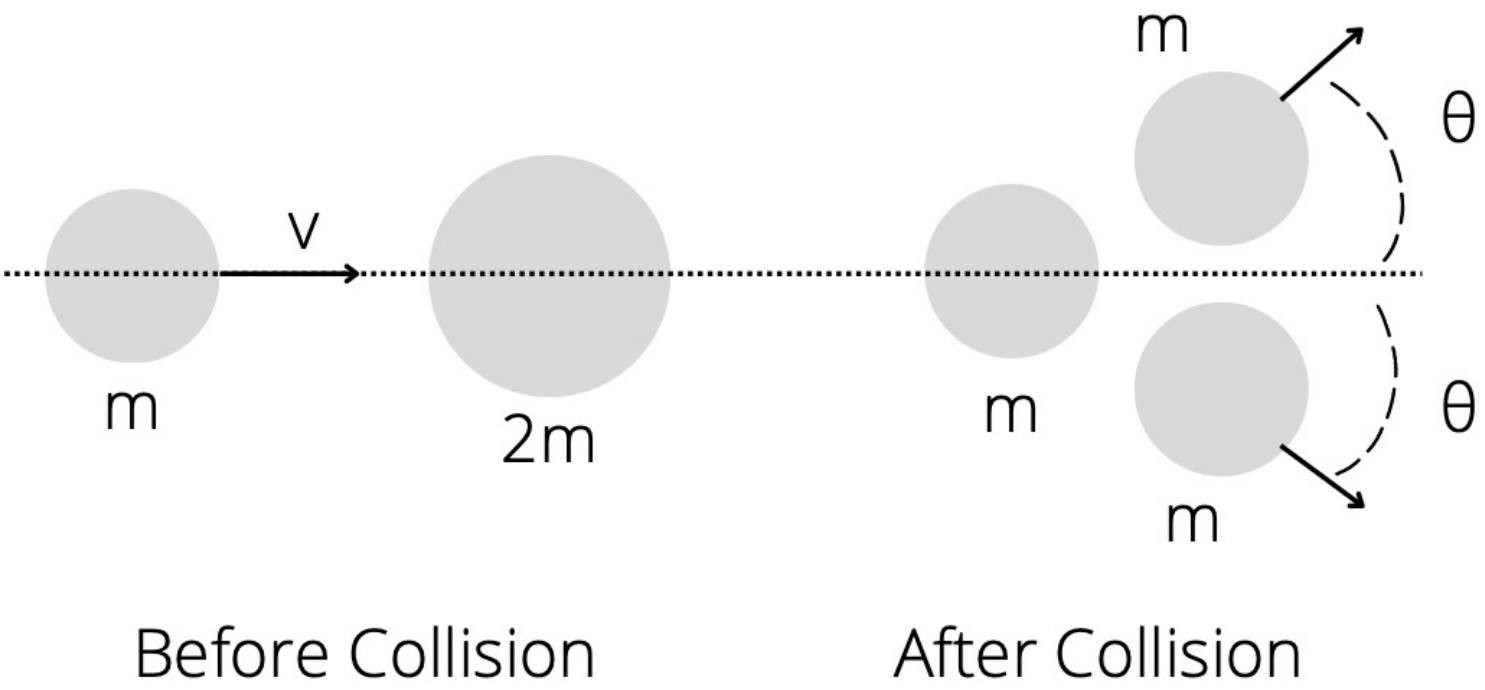
**(B)** will never be attained

**(C)** $\sqrt{\frac{m}{12k}}v$

**(D)** $\sqrt{\frac{m}{6k}}v$

**Q.9**

A particle of mass $m$ is moving along the $x$-axis with speed $v$ when it collides with a particle of mass $2m$ initially at rest. After the collision, the first particle has come to rest and the second particle has split into two equal-mass pieces that are shown in the figure. Which of the following statement correctly describes the speeds of the two places?




**(A)** Each piece moves with speed $v$

**(B)** Each piece moves with speed $v/2$

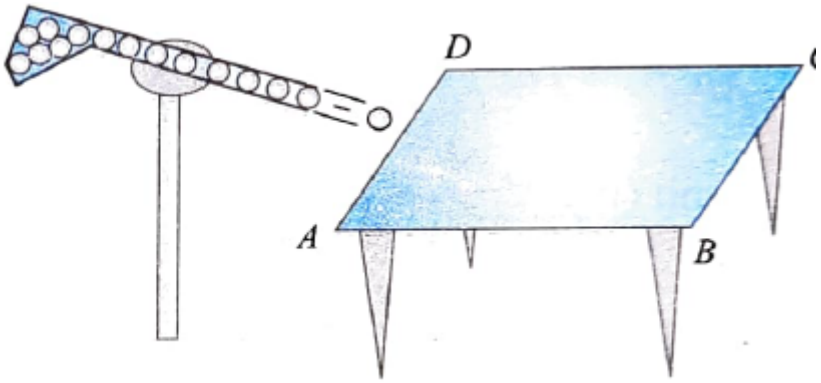
**(C)** One of the pieces moves with speed $v/2$, the other move with speed greater than $v/2$

**(D)** Each piece moves with speed greater than $v/2$

**Q.10**

A gun which fires small balls of mass $20\ \text{g}$ is firing $20\text{balls}$ per second on the smooth horizontal table surface $ABCD$ the collision is perfectly elastic and balls are striking at the centre of table with a speed of $5\ \text{m/s}$ at an angle of $60^\circ$ with the vertical just before collision, then force exerted by one of the legs on ground is (assume total weight of the table is $0.2\ \text{kg}$)

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**(A)** $0.5$ N

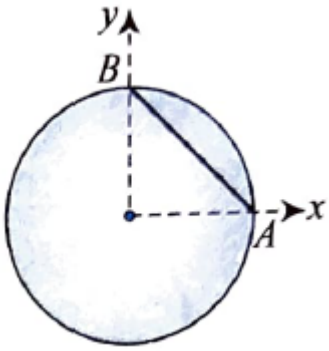
**(B)** $1$ N

**(C)** $0.25$ N

**(D)** $0.75$ N

**Q.11**

An object comprises of a uniform ring of radius $R$ and its uniform chord $AB$ (not necessarily made of the same material) as shown. Which of the following can not be the centre of mass of the object?




**(A)** $\left(\frac{R}{3}, \frac{R}{3}\right)$

**(B)** $\left(\frac{R}{3}, \frac{R}{2}\right)$

**(C)** $\left(\frac{R}{4}, \frac{R}{4}\right)$

**(D)** None of these

**Q.12**

A ball of mass $m$ moving with speed $u$ undergoes a head-on elastic collision with a ball of mass $nm$ initially at rest. The fraction of the incident energy transferred to the second ball is

**(A)** $\frac{n}{1+n}$

**(B)** $\frac{n}{(1+n)^2}$

**(C)** $\frac{2n}{(1+n)^2}$

**(D)** $\frac{4n}{(1+n)^2}$

**Q.13**

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An object initially at rest explodes into three fragments $A$ $B$ and $C$. The momentum of $A$ is $p\hat{i}$ and that of $B$ is $\sqrt{3}p\hat{j}$ where $p$ is a $+$ ve number. The momentum of $C$ will be

**(A)** $(1+\sqrt{3})p$ in a direction making angle $120^\circ$ with that of $A$

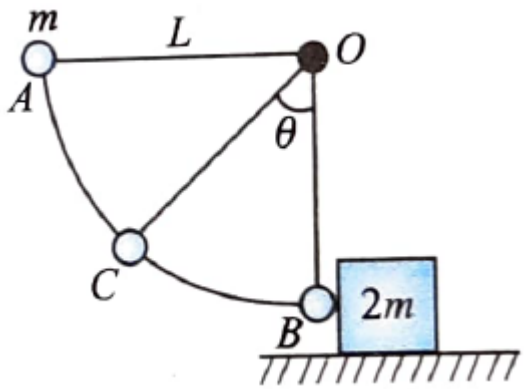
**(B)** $(1+\sqrt{3})p$ in a direction making angle $150^\circ$ with that of $B$

**(C)** $2p$ in a direction making angle $150^\circ$ with that of $A$

**(D)** $2p$ in a direction making angle $150^\circ$ with that of $B$

**Q.14**

A ball of mass $m$ is attached to a cord of length $L$, pivoted at point $O$, as shown in figure. The ball is released from rest at point $A$, swings down and makes an inelastic collision with a block of mass $2m$ kept on a rough horizontal floor. The coefficient of restitution of collision is $e = 2/3$ and coefficient of friction between block and surface is $\mu$. After collision, the ball comes momentarily to rest at $C$ when cord makes an angle of $\theta$ with the vertical and block moves a distance of $3L/2$ on rough horizontal floor before stopping. The values of $\mu$ and $\theta$ are, respectively,




**(A)** $\frac{50}{243}, \cos^{-1}\left(\frac{80}{81}\right)$

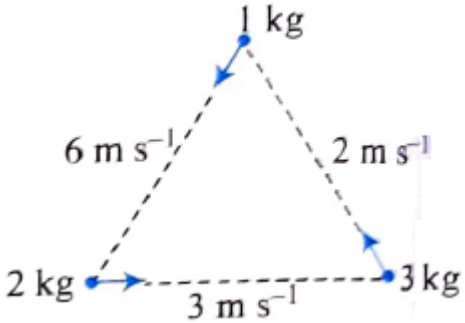
**(B)** $\frac{50}{81}, \cos^{-1}\left(\frac{80}{81}\right)$

**(C)** $\frac{2}{81}, \cos^{-1}\left(\frac{80}{243}\right)$

**(D)** $\frac{2}{243}, \cos^{-1}\left(\frac{80}{243}\right)$

**Q.15**

Three particles of masses $1\ \text{kg}, 2\ \text{kg}$ and $3\ \text{kg}$ are situated a the corners of an equilateral triangle move at speed $6\ \text{ms}^{-1}$ $3\ \text{ms}^{-1}$ and $2\ \text{ms}^{-1}$ respectively. Each particle maintains a direction towards the particle at the next corner symmetrically. Find velocity of $\text{CM}$ of the system at this instant.




**(A)** $3\ \text{ms}^{-1}$

**(B)** $5\ \text{ms}^{-1}$

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**(C)** $6\ \text{ms}^{-1}$

**(D)** zero

**Q.16**

A body of mass $1\ \text{kg}$ initially at rest, explodes and breaks into three fragments of masses in the ratio $1:1:3$. The two pieces of equal mass fly off perpendicular to each other with a speed of $30\ \text{ms}^{-1}$ each. What is the velocity of the heavier fragment?

**(A)** $10\sqrt{2}\ \text{m s}^{-1}$

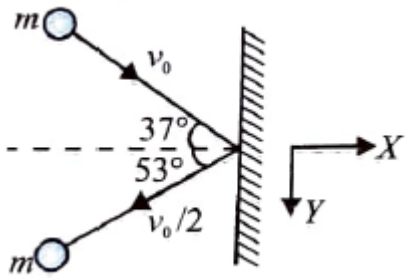
**(B)** $5\sqrt{3}\ \text{m s}^{-1}$

**(C)** $10\sqrt{3}\ \text{ms}^{-1}$

**(D)** $5\sqrt{2}\ \text{ms}^{-1}$

**Q.17**

A ball of mass $m$ moving with velocity $v_0$ collides with a wall as shown in figure. After impact it rebounds with a velocity $v_0/2$. The component of impulse acting on the ball along the wall is




**(A)** $\frac{mv_0}{2}\hat{j}$

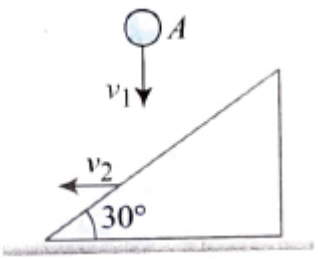
**(B)** $-\frac{mv_0}{2}\hat{j}$

**(C)** $-\frac{mv_0}{5}\hat{j}$

**(D)** none of these

**Q.18**

A ball $A$ is falling vertically downwards with velocity $v_1$. It strikes elastically with a wedge moving horizontally with velocity $v_2$ as shown in figure. What must be the ratio $v_1/v_2$ so that the ball bounces back in vertically upwards direction relative to wedge?




**(A)** $\sqrt{3}$

**(B)** $\frac{1}{\sqrt{3}}$

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**(C)** 2

**(D)** $\frac{1}{2}$

**Q.19**

A body falling vertically downwards under gravity breaks in two parts of unequal masses. The centre of mass of the two parts taken together shifts horizontally towards:

**(A)** heavier piece

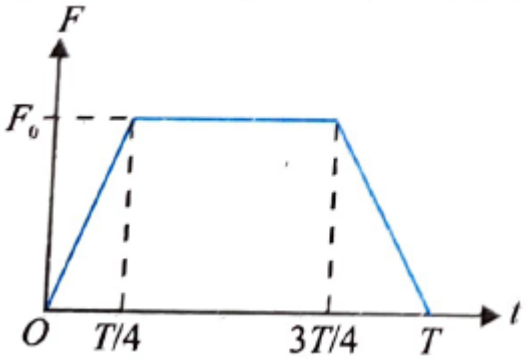
**(B)** lighter piece

**(C)** does not shift horizontally

**(D)** depends on the vertical velocity at the time of breaking

**Q.20**

A particle of mass $m$ moving with a velocity $v$ makes an elastic one dimensional collision with a stationary particle of mass $m$ establishing a contact with it for extremely small time $T$. Their force of contact increases from zero to $F_0$ linearly in time $T/4$, remains constant for a further time $T/2$ and decreases linearly from $F_0$ to zero in further time $T/4$ as shown in figure. The magnitude possessed by $F_0$ is




**(A)** $\frac{mu}{T}$

**(B)** $\frac{2mu}{T}$

**(C)** $\frac{4mu}{3T}$

**(D)** $\frac{3mu}{4T}$

**Q.21**

A man of mass $M = 58\ \text{kg}$ jumps from an aeroplane as shown in figure. He sees the hard ground below him and a lake at a distance $d = 1\ \text{m}$ from the point directly below him. He immediately puts off his jacket (mass $m = 2\ \text{kg}$ ) and throws it in a direction directly away from the lake. If he just fails to strike the ground, find the distance (in $\text{m}$) he should walk now to pick his jacket. (Neglect air resistance and take the velocity of man at the time of jump with respect to earth zero.)

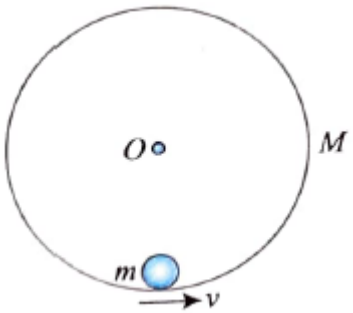



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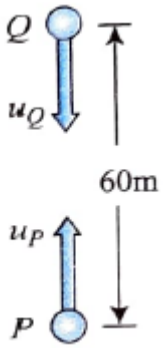
**Q.22**

A ring of mass $M = 90$ gram and radius $R = 3$ meter is kept on a frictionless horizontal surface such that its plane is parallel to horizontal plane. A particle of mass $m = 10$ gram is placed in contact with the inner surface of ring as shown figure. An initial velocity $v = \sqrt{2}\ \mathrm{m/s}$ is given to the particle along the tangent of the ring. Find the magnitude of the force interaction (in milli-newton) between the particle and ring.



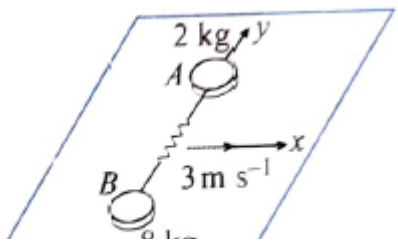

**Q.23**

Two particles $P$ and $Q$ have mass $4.0\ \mathrm{kg}$ and $1\ \mathrm{kg}$ respectively They are projected along a vertical line with velocity $u_p\ 20\ \mathrm{m/s}$ and $u_Q = 5\ \mathrm{m/s}$ when separation between them is $\mathrm{w}_2\ 100\ \mathrm{m}$. $P$ was projected vertically up while $Q$ was projected vertically down. Calculate the maximum height attained (in $\mathrm{m}$ ) by the centre of mass of the system of two particles measured from the initial position of $P$. Assume that the particles do not collide and that the ground is far below their point of projection $\left[g = 10\ \mathrm{m/s^2}\right]$ (Use the intregral part only)

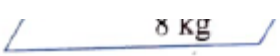



**Q.24**

A massless spring of force constant $1000\mathrm{Nm}^{-1}$ is compressed a distance of $20\ \mathrm{cm}$ between discs of $8\ \mathrm{kg}$ and $2\ \mathrm{kg}$, spring is not attached to discs. The system is given an initial velocity $3\ \mathrm{ms}^{-1}$ perpendicular to length of spring as shown in the figure. What is ground frame velocity of $2\ \mathrm{kg}$ block (in $\mathrm{ms}^{-1}$) when spring regains its natural length.

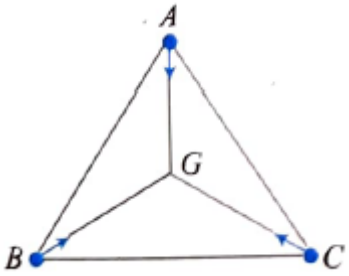



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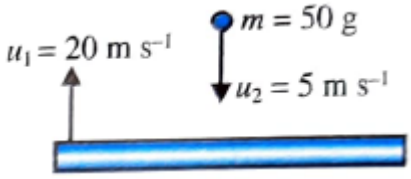

**Q.25**

Three particles $A, B$ and $C$ of equal mass move with equal speed $V = 5\ \text{ms}^{-1}$ along the medians of an equilateral triangle as shown in the figure. They collide at the centroid $G$ of the triangle. After the collision, $A$ comes to rest, $B$ retraces its path with the speed $V = 5\ \text{ms}^{-1}$. What is the speed of $C$ ?



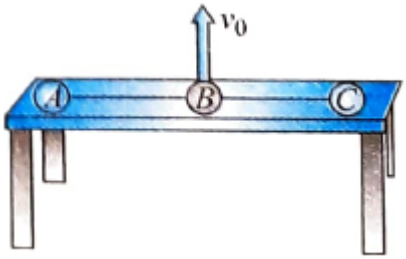

**Q.26**

An elevator platform is going up at a speed $20\ \text{ms}^{-1}$ and during its upward motion a small ball of $50\ \text{g}$ mass falling in downward direction strikes the platform elastically at a speed $5\ \text{ms}^{-1}$. Find the speed (in $\text{ms}^{-1}$) with which the ball rebounds.



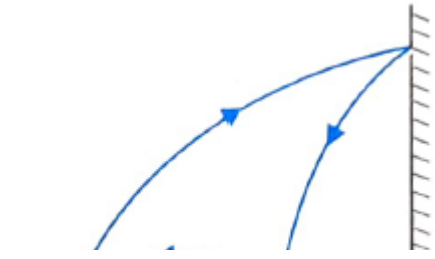

**Q.27**

Three identical particles are placed on a horizontal smooth table, connected with strings as shown. The particle is imparted a velocity $v_0 = 9\ \text{m/s}$ in horizontal direction perpendicular to the line $ABC$. The speed (in $\text{m/s}$) of particle $A$ when it is about to collide with $C$.

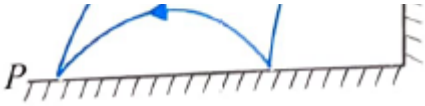



**Q.28**

A small ball is projected from point $P$ towards a vertical wall as shown in figure. It hits the wall when its velocity is horizontal. Ball reaches point $P$ after one bounce on the floor. The coefficient of restitution assuming it to be same for two collisions is $n/2$. All surfaces are smooth. Find the value of $n$.

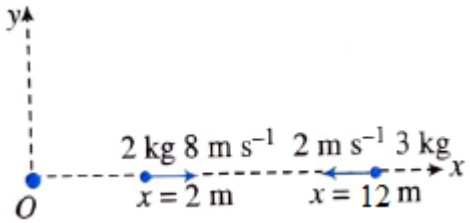


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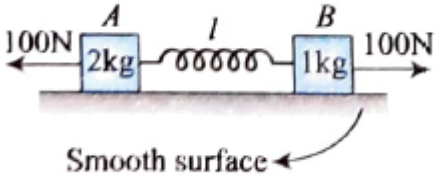

**Q.29**

Figure shows position and velocities of two particles moving under mutual gravitational attraction in space at time $t = 0$. The position of centre of mass after one second is '*'m. Fill '*'.




**Q.30**

In the figure shown initially spring is in relaxed state and blocks are at rest. Now $100\ \text{N}$ force is applied on block $A$ and $B$ as shown in figure. After some time velocity of '$A$' becomes $2\ \text{m/s}$ and that of ' $B'4\ \text{m/s}$ and block $A$ displaced by amount $10\ \text{cm}$ towards left and spring is stretched by amount $30\ \text{cm}$. Then work done by spring force on $A$ is $W_0$ joule. Find $|W_0|$ (in J).




## Answers & Solutions

**Q.1 Answer:**

$1, 3, 2$

**Solution:**

**In 1 - > small portion from top left corner is removed there will be shift in x-co ordinate of COM in right direction**

**In 2 - > small portion from top left corner and bottom left corner also are removed there will be shift in x-co ordinate of COM in right direction (greater shift in right direction than 1)**

**In 3 - > small portion from top left corner and bottom left corner also are removed there will be shift in x-co ordinate of $\text{COM}$ in right direction (greater shift in right direction than 1 ) and one portion from right top corner is than removed so a shift in leftward direction is attained which is comparable to 1. but is still in rightward direction to $1$.**

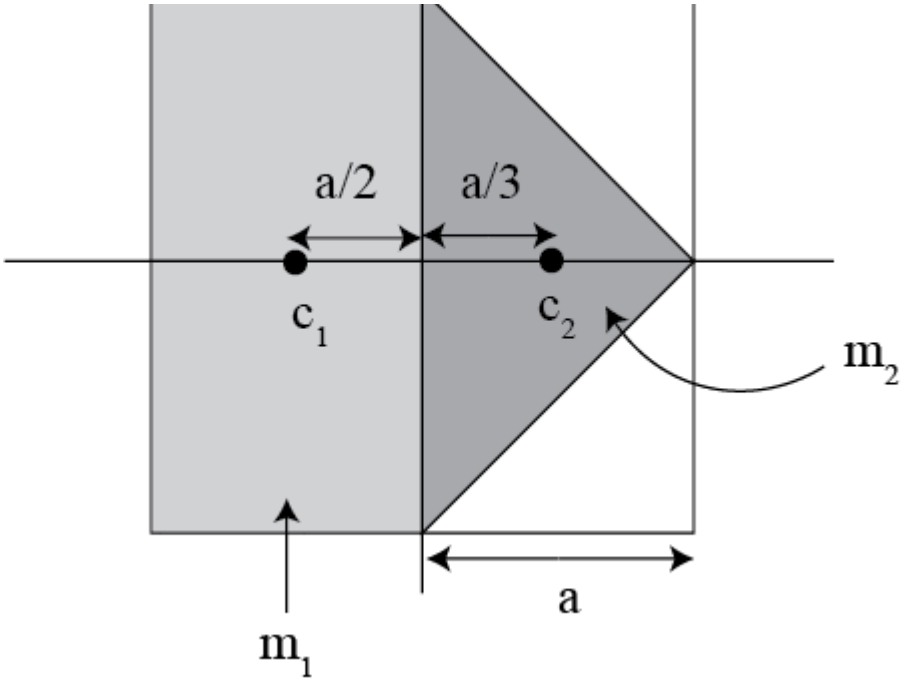
**Q.2 Answer:**

$\left(-\frac{a}{12}, 0\right)$

**Solution:**



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**let, c.o.m of $m_1$ be $c_1$ $\left(-\frac{a}{2}, 0\right)$**
**Ee com of $\mathrm{M}_2$ be $\mathrm{C}_2$ $\left(\frac{a}{3}, 0\right)$**
**$\left[\because\right.$ com of a sa reat is at its centre $G$ com of $a\Delta$ is at $\left(\frac{h}{3}, 0\right)$**
**so, $\bar{x}$ of full page $= \frac{m_1(-a/2)+m_2(1/3)}{m}$**
$= \frac{\frac{m}{2}(-a/2)+\frac{m}{2}a/3}{m}$ $\left[\begin{array}{l}\because m_1 = m_2 \\ = \frac{m}{2}\end{array}\right]$
$= \frac{-a}{4} + \frac{9}{6}$
$= \frac{-2}{24}a = -\frac{9}{12} \cdot$ Ans 12

**Q.3 Answer:**

$\frac{L}{4}$

**Solution:**

**If the force is applied to the center of mass of the whole system then the system will perform only transitional motion not rotational motion .**

**For COM:**

$$\mathrm{X_{COM}} = \frac{\mathrm{x_1\ m_1 + x_2\ m_2}}{\mathrm{m_1+m_2}}$$
$$\mathrm{x_{COM}} = \frac{0.\mathrm{M}+\mathrm{M}\times\frac{\mathrm{L}}{2}}{\mathrm{M+M}}$$
$$\mathrm{X_{COM}} = \frac{\mathrm{L}}{4}$$

**Q.4 Answer:**

4.5 cm

**Solution:**

$$\mathrm{Y_{CM}} = \frac{\mathrm{m_1 x_1 + m_2 x_2}}{\mathrm{m_1+m_2}}$$
$$0 = \frac{\rho\pi(10.5)^2\mathrm{t}\times 3.5 + \rho\pi\left(14^2-10.5^2\right)\mathrm{t x_2}}{\mathrm{m_1+m_2}}$$

**Here $\pi \mathrm{r}^2\mathrm{t}$ is volume and mass is written in terms of density times volume, where $\rho$ is the density of plate**

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**material and $t$ is the thickness of the plate.**

$x_2 = \frac{10.5^2 \times 3.5}{14^2 - 10.5^2}$

$x_2 = 4.5 \text{ cm}$

**Q.5 Answer:**

initially accelerate with $\vec{g}$ and then with accelerate with $g/9$

**Solution:**

**Initially, both the blocks are under the gravity effect, so**

$a_{CM} = \frac{4\,g + 2\,g}{4+2} = g$

**But once the jerk in the string occurs and string becomes taut, the $4$ kg block moves down with acceleration $g/3$ and $2$ kg block moves up with acceleration g / 3. So,** $a_{CM} = \frac{4 \times \frac{g}{3} - \frac{2\,g}{3}}{6} = \frac{g}{9}$

**Q.6 Answer:**

$6\sqrt{2}$ N s

**Solution:**

**Initial velocity of the particle** $\overrightarrow{v}_1 = 3\hat{i} m/s$

**Final velocity of the particle** $\overrightarrow{v_2} = 3\hat{j} m/s$

**Change in momentum** $\Delta\overrightarrow{P} = m\left(\overrightarrow{v}_2 - \overrightarrow{v}_1\right) = 2 \times 3(\hat{j} - \hat{i})$

$|\Delta\overrightarrow{P}| = 2 \times 3\sqrt{2} = 6\sqrt{2} \text{ N s}$

**Magnitude of impulse** $I = |\Delta\overrightarrow{P}| = 6\sqrt{2} \text{ N s}$

**Q.7 Answer:**

$6E_0$

**Solution:**

**Let mass of each fragment be** $m$.

**Kinetic energy of each fragment** $E_1 = E_2 = E_3 = E_0$.

**Thus magnitude of momentum of each fragment** $p_1 = p_2 = p_3 = \sqrt{2mE_0}$.

**Fourth fragment will fly in a direction opposite to the direction of resultant of** $p_1, p_2$ **and** $p_3$ **so that total momentum of the system after explosion also remains zero.**

**Momentum of the fourth fragment** $p_4 = \sqrt{p_1^2 + p_2^2 + p_3^2}$

$\therefore p_4 = \sqrt{2mE_o + 2mE_o + 2mE_o} = \sqrt{6mE_o}$

**So, kinetic energy of fourth fragment** $E_4 = \frac{p_4^2}{2\,m} = \frac{6mE_o}{2\,m} = 3E_o$

**Energy of explosion** $E = E_1 + E_2 + E_3 + E_4$

$\therefore E = E_o + E_o + E_o + 3E_o = 6E_o$

**Q.8 Answer:**

$\sqrt{\frac{m}{12k}}v$

**Solution:**

**mass of block A** $= m$

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**mass of block** $R = 2\ \mathrm{m}$

**Spring force constant** $=, k,$

**Velocity of masses** $= V$

**mass of block** $c = m$

**First consider the inelastic collision between** $B$ **and** $C,$

$2mV = (m + 2m)V_B$

$2mV = 3mV_B$

$\frac{2V}{3} = V_C \Rightarrow$ **velocity of** $B$ **and** $C$

**The velocity of center of mass during maximum compression**

$V_{cm} = \frac{3\ \mathrm{mv}}{4m}.$

$(2 + 1 + 1)V_{cm} = 3\mathrm{mV}$

$\mathrm{m}4V_{\mathrm{cm}} = 3\mathrm{mV} \rightarrow \frac{3\ \mathrm{V}}{4} = V_{cm}$

**So, By conservation of energy,**

$\frac{1}{2}mv_1^2 + \frac{1}{23}mv^2(B + c) = \frac{1}{24}mv_{cm}^2 + \frac{1}{2}kx^2$

**By substituiting, we get.** $\Rightarrow \frac{1}{24}m^2 = \frac{1}{2}kx^2 \Rightarrow \frac{m}{12}v^2 = x^2$

$x = \sqrt{\frac{mv^2}{nk}} = \sqrt{\frac{m}{12k}}V$

$= x\sqrt{\frac{m}{12k}}V$

**Q.9 Answer:**

Each piece moves with speed greater than $v/2$

**Solution:**

**Let** $\mathrm{V}'_{\mathrm{A}}$ **and** $\mathrm{V}'_{\mathrm{B}}$ **be the velocity of** $\mathrm{A}$ **and** $\mathrm{B}$ **after collision. From the conservation of momentum,**

$\mathrm{M_A V_A} + \mathrm{M_B V_B} = \mathrm{M_A V'_A} + \quad \mathrm{M'_B V'_B}$

**But,** $\mathrm{M_B} = 2\ \mathrm{m}$ **and** $\mathrm{M_A} = \mathrm{m}$

**From** $(1),$ $=> \mathrm{mv}^{+}0 = 0 + 2\left(\frac{1}{2} \times 2\ \mathrm{m}\right)\mathrm{v}'\ \mathrm{Cos}\,\theta$

$\Rightarrow \mathrm{v} = 2\mathrm{v}'\ \mathrm{Cos}\,\theta$

$\mathrm{v}' = \frac{\mathrm{v}}{2\ \mathrm{Cos}\,\theta} = \frac{1}{\mathrm{Cos}\,\theta}\left(\frac{\mathrm{v}}{2}\right)$

**Hence each piece moves with speed greater than** $\frac{\mathrm{v}}{2}$

**Q.10 Answer:**

$1\ \mathrm{N}$

**Solution:**

**Mass of the balls supplied per second** $= 20 \times 0.02\ \mathrm{kg} = 0.4\ \mathrm{kg/s}$

**Velocity in the vertical direction before collision=Velocity in vertical direction after collision** $= 5 \times \cos 60^\circ = 2.5\ \mathrm{m/s}$

**Thus change in velocity in each collision** $= 5\ \mathrm{m/s}$

**Thus the change in momentum per unit time= Force exerted on the ground due to collision** $= \frac{\mathrm{dp}}{\mathrm{dt}} = \frac{\mathrm{dm}}{\mathrm{dt}}\Delta\mathrm{v} = 0.4\ \mathrm{kg/s} \times 5\ \mathrm{m/s} = 2\ \mathrm{N}$

**Thus force exerted on each leg** $= \frac{1}{4} \times (2\ \mathrm{N} + (0.2 \times 10)\mathrm{N}) = 1\ \mathrm{N}$

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**Q.11 Answer:**

$\left(\frac{R}{3}, \frac{R}{2}\right)$

**Solution:**

**Since the chord is a uniform chord and the center of mass coordinate of system will be displaced in $x$ and y coordinate by same value. Hence $X$ and $Y$ coordinate will be same. So the wrong aption is $\left(\frac{R}{3}, \frac{R}{2}\right)$**

**Q.12 Answer:**

$\frac{4n}{(1+n)^2}$

**Solution:**

**Given:** $m_1 = m \quad m_2 = mm$

**Let the velocities of $A$ and $B$ after the collision be $v_1$ and $v_2$ respectively. initial velocity of $A$ before collision is $v$ and that of $B$ is zero. Initial kinetic energy of the system** $E = \frac{1}{2}mv^2$

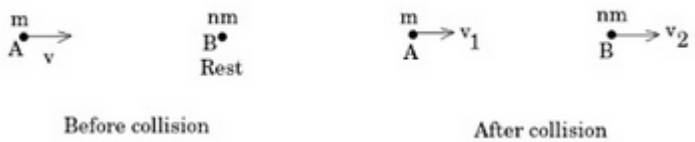
**Using** $v_2 = \frac{(m_2 - em_1)u_2 + (1+e)m_1u_1}{m_1+m_2}$

$v_2 = \frac{(nm - 1\times m)(0) + (1+1)m(v)}{m+nm} \implies v_2 = \frac{2v}{(1+n)}$

**Thus Kinetic energy of $B$ after the collision** $E'_B = \frac{1}{2}(nm)v_2^2$

$E'_B = \frac{1}{2}(nm)\frac{4v^2}{(1+n)^2}$

**Fraction of total kinetic energy retained by B** $\frac{E'_B}{E} = \frac{4n}{(n+1)^2}$




**Q.13 Answer:**

$2p$ in a direction making angle $150^\circ$ with that of $B$

**Solution:**

**Let the fragments $C$ moves with momentum $P_3$ making an angle $\theta$ with negative $x$ axis.**

**Initial momentum of the bomb is Zero i.e $P_i = 0$ Applying conservation of momentum in $x$ direction:**

$0 = p - P_3\cos\theta \implies P_3\cos\theta = p$ **(1)**

**Applying conservation of momentum in y direction:**

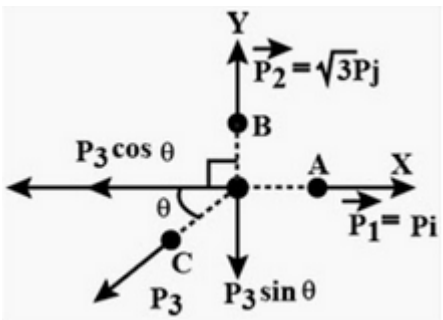
$0 = \sqrt{3}p - P_3\sin\theta \implies P_3\sin\theta = \sqrt{3}p$ **(2)**

**Squaring and adding (1) and (2),** $P_3^2 = p^2 + 3p^2$

$\implies P_3 = 2p$

**Now dividing (1) form (2) we get,** $\tan\theta = \sqrt{3} \implies \theta = 60^\circ$

**Angle made by $P_3$ with $P_2$** $\phi = 90^\circ + \theta = 90^\circ + 60^\circ = 150^\circ$




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**Q.14 Answer:**

$\frac{50}{243}, \cos^{-1}\left(\frac{80}{81}\right)$

**Solution:**

**Velcoity of the ball just befoe collision is** $\mathrm{v}_0 = \sqrt{2\mathrm{gL}}$

**Applying momentum conservation along horizontal direction(because momentum is conserved in collision along the line of impact), we get**

$$\mathrm{mv}_0 = -\mathrm{mv}_1 + 2\mathrm{mv}_2$$

**Applying coefficient of restitution equation, we get**

$$e = \frac{2}{3} = \frac{v_1 + v_2}{v_0}$$

**Solving the above two equations, we get**

$$\mathrm{v}_1 = \frac{\mathrm{v}_0}{9} \text{ and } \mathrm{v}_2 = \frac{5\mathrm{v}_0}{9}$$

**As the block moves a distance of** $3\ \mathrm{L}/2$ **before coming to rest, so from work-energy theorem,**

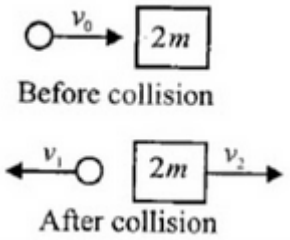
$$0 - \frac{1}{2}(2\ \mathrm{m})\mathrm{v}_2^2 = -\mu \times 2\mathrm{mg}\frac{3\ \mathrm{L}}{2} \Rightarrow \mathrm{v}_2^2 = 3\mu\mathrm{gL}$$

$$\frac{25}{81} \times 2\mathrm{gL} = 3\mu\mathrm{gL} \Rightarrow \mu = \frac{50}{243}$$

**For ball, KE is converted to gravitational potential energy after collision, so**

$$0 - \frac{\mathrm{mv}_1^2}{2} = -\mathrm{mg} \times \mathrm{L}(1 - \cos\theta)$$

$$\cos\theta = \frac{80}{81}$$




**Q.15 Answer:**

zero

**Solution:**

$$\text{Given :} \quad \mathrm{m}_1 = 1\ \mathrm{kg} \quad \mathrm{m}_2 = 2\ \mathrm{kg} \quad \mathrm{m}_3 = 3\ \mathrm{kg}$$

$$\mathrm{v}_1 = 6\ \mathrm{m/s} \quad \mathrm{v}_2 = 3\ \mathrm{m/s} \quad \mathrm{v}_3 = 2\ \mathrm{m/s}$$

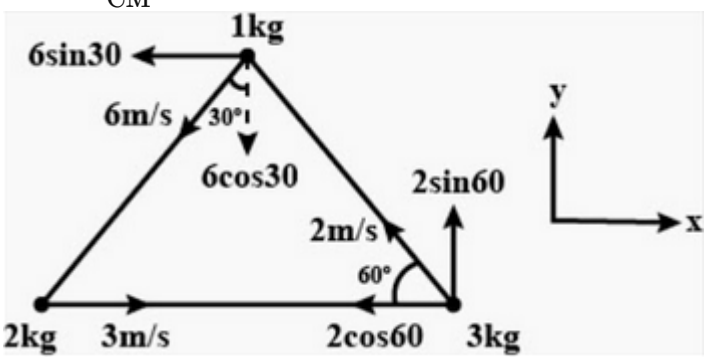
**Let velocity of their centre of mass be** $\mathrm{V}_{\mathrm{CM}}$

$$\text{Using} \quad \overrightarrow{V_{\mathrm{CM}}\mathrm{CM}} = \frac{\mathrm{m}_1\vec{\mathrm{v}}_1 + \mathrm{m}_2\vec{\mathrm{v}_2} + \mathrm{m}_3\vec{\mathrm{v}_3}}{\mathrm{m}_1 + \mathrm{m}_2 + \mathrm{m}_3}$$

$$\mathrm{V}_{\mathrm{CM}} = \frac{1\left(-6\sin 30^\circ\hat{\mathrm{i}} - 6\cos 30^\circ\hat{\mathrm{j}}\right) + 2(3\hat{\mathrm{i}}) + 3\left(-2\cos 60^\circ\hat{\mathrm{i}} + 2\sin 60^\circ\hat{\mathrm{j}}\right)}{1+2+3}$$

$$\overrightarrow{V_{\mathrm{CM}}} = \frac{1\left(-6\times\frac{1}{2}\hat{\mathrm{i}} - 6\times\frac{\sqrt{3}}{2}\hat{\mathrm{j}}\right) + 2(3\hat{\mathrm{i}}) + 3\left(-2\times\frac{1}{2}\hat{\mathrm{i}} + 2\times\frac{\sqrt{3}}{2}\hat{\mathrm{j}}\right)}{6}$$

$$\Longrightarrow \vec{\mathrm{V}_{\mathrm{CM}}^{\mathrm{i}}} = 0$$




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**Q.16 Answer:**

$10\sqrt{2}\ \mathrm{m\ s^{-1}}$

**Solution:**

**Total mass of the body** $\mathrm{M} = 1\ \mathrm{kg}$

**So, mass of first fragment** $\mathrm{m_1} = \frac{1}{5} \times 1 = 0.2\ \mathrm{kg}$

**Mass of second fragment** $\mathrm{m_2} = \frac{1}{5} \times 1 = 0.2\ \mathrm{kg}$

**Mass of third fragment** $\mathrm{m_3} = \frac{3}{5} \times 1 = 0.6\ \mathrm{kg}$

**Velocity of masses** $\mathrm{m_1}$ **and** $\mathrm{m_2}, \mathrm{v} = 30\ \mathrm{m/s}$

**Momentum of first fragment** $\mathrm{p_1} = \mathrm{m_1 v} = 0.2 \times 30 = 6\ \mathrm{kg\ m/s}$

**Similarly,** $\mathrm{p_2} = \mathrm{m_2 v} = 0.2 \times 30 = 6\ \mathrm{kg\ m/s}$

**Third fragment will fly in a direction opposite to the direction of resultant of** $\mathrm{p_1}$ **and** $\mathrm{p_2}$ **so that total momentum of the system after explosion remains zero.**

**Momentum of the third fragment** $\mathrm{p_3} = \sqrt{\mathrm{p_1^2 + p_2^2}}$

$\therefore \mathrm{p_3} = \sqrt{6^2 + 6^2} = 6\sqrt{2}\ \mathrm{kg\ m/s}$

**Thus speed of third fragment** $\mathrm{v_3} = \frac{\mathrm{p_3}}{\mathrm{m_3}} = \frac{6\sqrt{2}}{0.6} = 10\sqrt{2}\ \mathrm{m/s}$

$\Longrightarrow \mathrm{x} = 2$

**Q.17 Answer:**

$-\frac{mv_0}{5}\hat{j}$

**Solution:**

**Component of initial momentum along the wall is**

$\mathrm{P_{iy}} = \mathrm{mv_0} \sin 37^\circ \hat{\mathrm{j}}$

**Component of final momentum along the wall is**

$\mathrm{P_{fy}} = \frac{\mathrm{mv_0}}{2} \sin 53^\circ \hat{\mathrm{j}}$

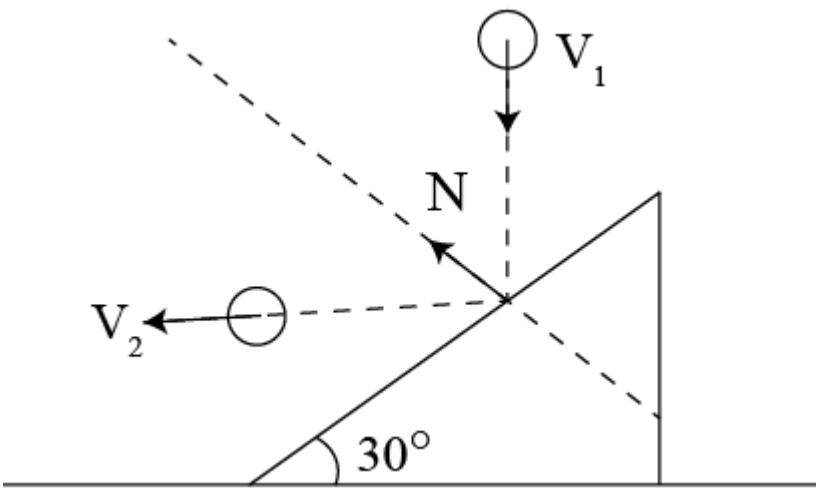
**So, component of impulse along the wall is** $\mathrm{J_y} = \mathrm{P_{fy}} - \mathrm{P_{iy}}$

**Q.18 Answer:**

$\frac{1}{\sqrt{3}}$

**Solution:**

**Given that a ball strike at weadge with velocity** $V_1$ **ond goes velocity** $v_2$




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**Force Apply only perpendicular from the weadge**

**So momentum range only in the direction of the momentum**

$p_i = p_t$

$mv_1 \cos 30^\circ = mv_2 \sin 30^\circ$

$\frac{v_1}{v_2} = \tan(30^\circ)$

$\frac{v_1}{v_2} = \frac{1}{\sqrt{3}}$

**Q.19 Answer:**

does not shift horizontally

**Solution:**

**Since the velocity is in the downwards direction the motion of the COM will also be in that particular direction.**

**As the horizontal component of force is zero so COM will not move horizontally.**

**Q.20 Answer:**

$\frac{4mu}{3T}$

**Solution:**

**Initial velocity of the moving body (A) is $v$ and that of body (B) is zero. The two colliding bodies having same mass exchange their respective velocities after the head-on elastic collision.**

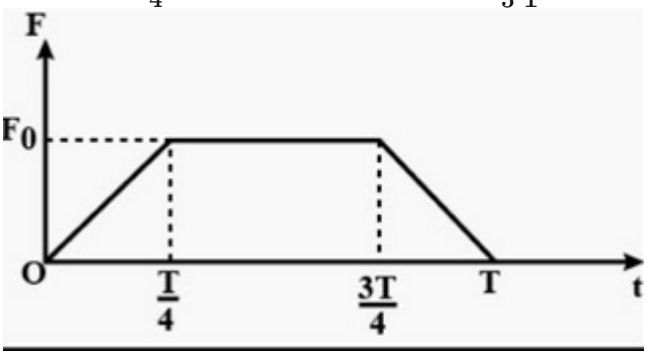
**Thus velocity of the body (A) after the collision is zero.**

$\Longrightarrow$ **Change in momentum of body (A),** $\Delta \mathrm{p} = \mathrm{mv}$

**Area under the graph shown,** $\mathrm{A} = \frac{1}{2}\,\mathrm{F}_0 \times \frac{\mathrm{T}}{4} + \mathrm{F}_0 \times \frac{\mathrm{T}}{2} + \frac{1}{2}\,\mathrm{F}_\mathrm{o} \times \frac{\mathrm{T}}{4} = \frac{3}{4}\,\mathrm{T}\,\mathrm{F}_\mathrm{o}$

**Using** $\Delta \mathrm{p} = \int \mathrm{F dt}$ **= Area under F-t graph**

$\therefore \mathrm{mv} = \frac{3}{4}\mathrm{TF}_\mathrm{o} \quad \Longrightarrow \quad \mathrm{F}_\mathrm{o} = \frac{4\mathrm{mv}}{3\,\mathrm{T}}$




**Q.21 Answer:**

—

**Solution:**

**To save himself, the man throws his jacket in opposite direction to the lake. According to momentum conservation, he himself gets a velocity in the direction of the lake. During the motion as gravity is the only external force on the system (man plus jacket), centre of mass will not be displaced horizontally. Thus, centre of mass of the system falls vertically and when the man falls in the lake, jacket falls at a point such that the centre of mass of the man and the jacket will be directly below the point from where the**

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**man jumps).**

**As it is given that man falls at a distance $d$ from this point, it implies that jacket will fall at a distance $x$ in the opposite direction such that**

$mx = Md \Rightarrow x = \frac{M}{m}d = 29 \text{ m}$

**man has to travel a distance $x + d = 29 + 1 = 30$ m to pick his jacket.**

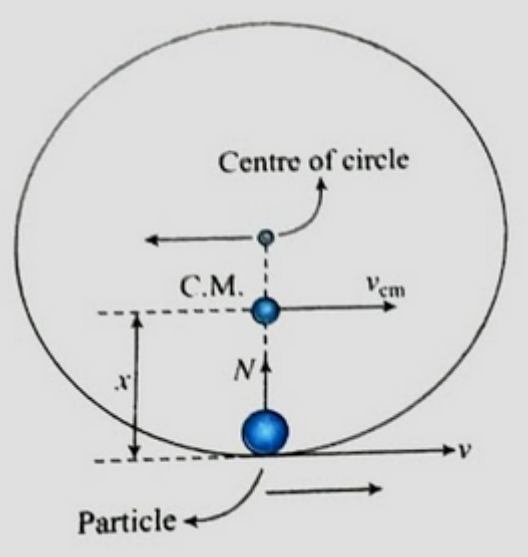
**Q.22 Answer:**

—

**Solution:**

**There is no external force acting on the system 'ring ' particle', it means the C.M. will move with constant velocity and centre of ring and particle will rotate about C'M.**

**The radius of rotation of the particle, $x = \frac{MR}{(M+m)}$**




**The normal reaction between the particle and ring provides required centripetal force for circular motion of the particle about C.M.**

**Hence**

$$N = \frac{m\left(v - v_{\text{cm}}\right)^2}{x}$$

$$= \frac{m\left[v - \frac{mv}{M+m}\right]^2}{\frac{MR}{(M+m)}} = \frac{mMv^2}{(m+M)}$$

**Substituting the value of $m$, $M$ and $v$ we get,**

$N = 6 \times 10^{-3}$ N

**Q.23 Answer:**

—

**Solution:**

$a_{\text{cm}} = \frac{F_{\text{ext}}}{\text{mass}} = \frac{m_P g + m_Q g}{m_P + m_Q} = g(\downarrow)$

$u_{\text{cm}} = \frac{m_P u_P + m_Q u_Q}{m_P + m_Q} = \frac{4\times 20 - 1\times 5}{4+1} = 15 \text{ m/s}(\uparrow)$

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**Initial height of COM above $P$ is** $h_0 = \frac{4\times0+1\times100}{5} = 20$ m

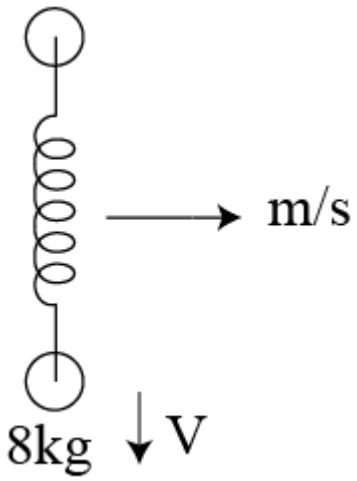
**Further height attained** $h = \frac{u_{cm}^2}{2g} = \frac{(15)^2}{2\times10} = 11.75$ m

**Total height** $H = 20 + 11.75 = 31.75$ m.

**Q.24 Answer:**

—

**Solution:**




**velocity of** $2$ kg **at Natural length** $u$

**conservation of momentum** $m, u = m_2 v$

$\Rightarrow \quad 24u = 8v$

$\Rightarrow \quad u = 4v$

**conservation of energy,**

$\frac{1}{2}kx^2 = \frac{1}{2}m_1 4^2 + \frac{1}{2}m_2 v^2$

$\Rightarrow \frac{1}{2} \times 1000 \times \frac{26}{100} \times \frac{20}{100} = \frac{1}{2} \times 24^2 + \frac{1}{2} \times 8 \times \left(\frac{4}{4}\right)^2$

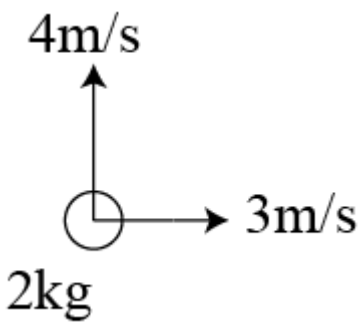
$\Rightarrow \quad 20 = u^2 + \frac{1}{2} \times 8 \times \frac{u^2}{16}$

$\Rightarrow \quad 20 = u^2 + \frac{u^2}{4}$

$\Rightarrow 20 = \frac{54^2}{9} \Rightarrow u^2 = \frac{46\times4}{5} \Rightarrow 4 = 4m$

$|\vec{v}| = \sqrt{4^2 + 3^2}$

$= 5$ m/s



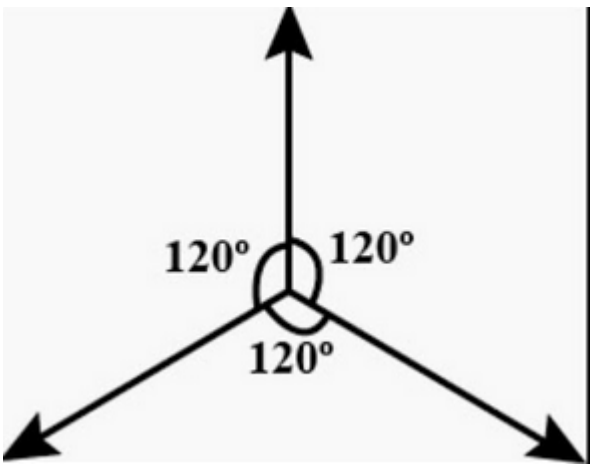

**Q.25 Answer:**

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**Solution:**

**The velocities of three particles are directed as shown in figure It is clear that the three velocity vectors are inclined at an angle of $12^{\circ}$. We know that the resultant of three equal vectors inclined at angle of $120^{\circ}$ is zero. So, the resultant initial momentum of the system is zero. It means that the resultant final momentum of the system must be zero. As particle A comes at rest after collision, the momentum of particle C must be equal and opposite to the momentum of particle $\mathrm{B}$, for the final momentum to be zero. Thus, the velocity of $\mathrm{C}$ should be equal to $\mathrm{v}$ and directed opposite to that of $\mathrm{B}$.**




**Q.26 Answer:**

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**Solution:**

**Given : Velocity of elevator** $\mathrm{v_e} = 20\ \mathrm{ms^{-1}}$ **(Upwards)**

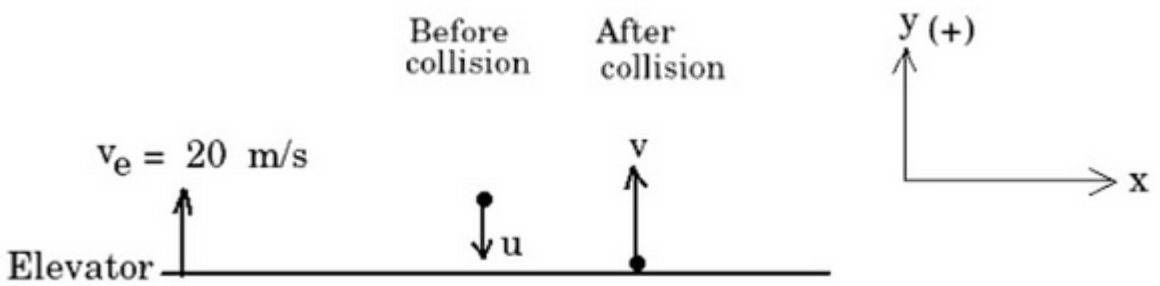
**Velocity of ball before collision** $\mathrm{u} = 5\ \mathrm{ms^{-1}}$ **(Downwards)**

**Let the velocity of ball after the collision be** $\mathrm{v}$

**Using Newton's Law of collision,** $\frac{\mathrm{v}-\mathrm{v_e}}{-\mathrm{u}-\mathrm{v_e}} = -\mathrm{e}$

$\frac{\mathrm{v}-20}{-5-20} = -1 \quad (\because \mathrm{e} = 1)$

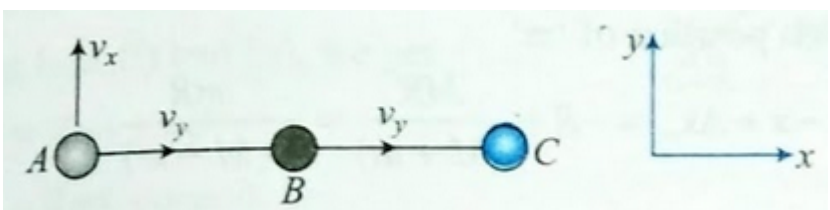
$\Longrightarrow \mathrm{v} = 45\ \mathrm{ms^{-1}}$




**Q.27 Answer:**

—

**Solution:**




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**Particle $A$ and $C$ will collide when the strings $BA$ and $BC$ become parallel to $y$ axis. At this time $B$ will be moving along. with velocity $v_y$ (say).**

**Component of velocity of $A\&C$ along $y$ axis will also be $v$, Particle $A$ has a component of velocity along $x$ axis as well (say $v_x$).**

**Momentum conservation along $y$ direction gives**

$$mv_0 = 3mv_y$$
$$\Rightarrow v_y = \frac{v_0}{3} = 3 \text{ m/s}$$

**Energy conservation,**

$$\frac{1}{2}mv_0^2 = \frac{1}{2}m\left(\frac{v_0}{3}\right)^2 + \frac{1}{2}(2m)\left(\left(\frac{v_0}{3}\right)^2 + v_x^2\right)$$

$$\left[\therefore \text{ Speed of } A \text{ and } C = \sqrt{v_y^2 + v_y^2} = \sqrt{\left(\frac{v_0}{3}\right)^2 + v_x^2}\right]$$

**Solving,** $v_x = 3\sqrt{3} \text{ m/s}$

**$\therefore$ Speed of $A$,** $v_A = \sqrt{(3\sqrt{3})^2 + 3^2} = 6 \text{ m/s}$

**Q.28 Answer:**

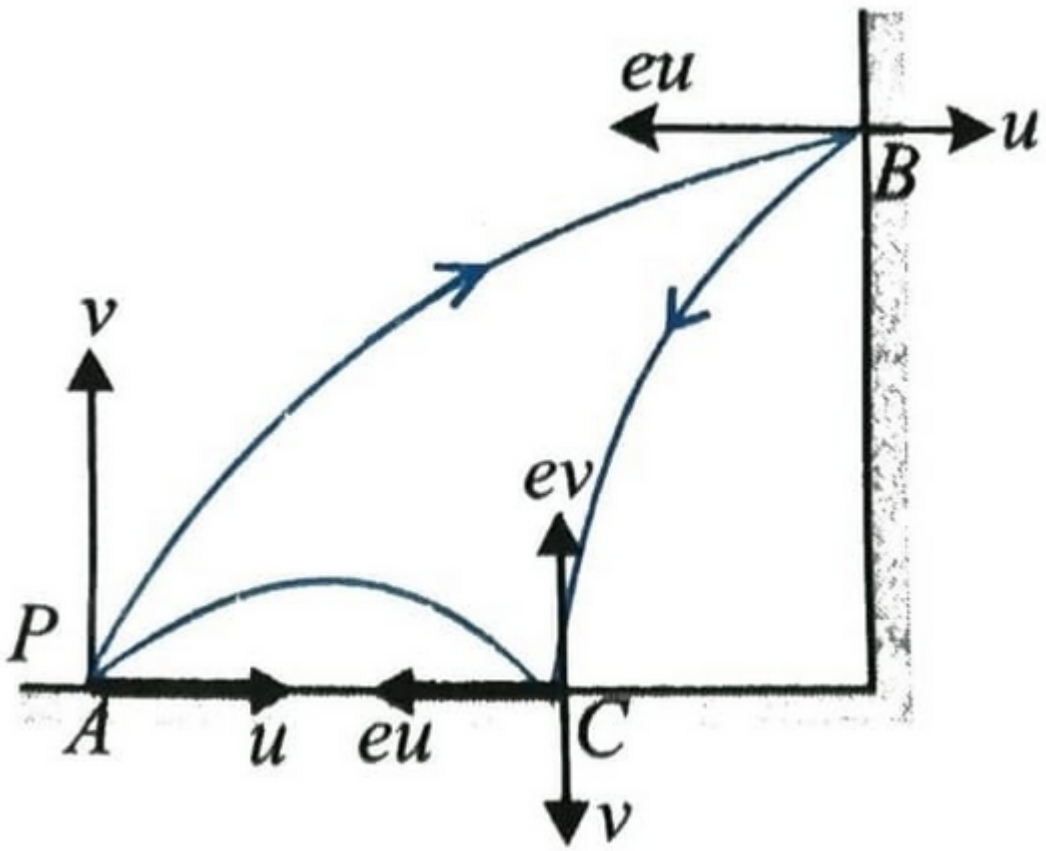
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**Solution:**

$u$ **time** $(AB) = eu($ **time** $BC) + eu($ **time** $CA)$

$$u\frac{v}{g} = eu\left(\frac{v}{g}\right) + eu \times 2\frac{ev}{g} \Rightarrow e = \frac{1}{2}$$

$$n = 1$$




**Q.29 Answer:**

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**Solution:**

**mutual force with not affect motion**

**the motion of center of mass can be supposed to be uniform**

**So initial velocity of center of mass**

$$\mathrm{u}=\frac{\mathrm{m_1v_1+m_2v_2}}{\mathrm{m_1+m_2}}=\frac{2\ \mathrm{kg}(8)+3\ \mathrm{kg}(-2)}{2+3}$$

$$\Rightarrow \mathrm{u}=\frac{10}{5}=2\ \mathrm{m/s} \quad +\mathrm{ve}$$

**towards increasing** $\mathrm{x}$

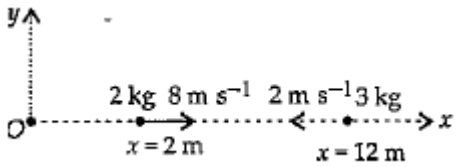
**distance in second will be** $x=xt=2x=2m$

$\Rightarrow$ **New position** $=$ **old position** $+2$

**or new position** $=\frac{\mathrm{m_1x_1+m_2x_2}}{\mathrm{m_1+m_2}}+2$

$$\mathrm{x}=\frac{2\times2+3\times12}{5}+2$$

$$\mathrm{x}=10 \text{ meters}$$




**Q.30 Answer:**

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**Solution:**

$$W_{\text{spring}}+W_{100\ \mathrm{N}}=\Delta K(\text{ on } A \text{ and } B)$$

$$W_{\text{spring}}+(100)\left(\frac{10}{100}\right)=\frac{1}{2}(2)(2)^2$$

$$W_{\text{spring}}=4-10=-6\ \mathrm{J}$$

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